DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 7: Apply Naive Bayes and Decision Tree Classification on a Dataset

Aim: To apply Naive Bayes and J48 Decision Tree classification algorithms using WEKA and compare their classification performance on a placement prediction dataset.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the concept of classification in data mining.
2. Understand supervised learning.
3. Apply the Naive Bayes classifier using WEKA.
4. Apply the J48 Decision Tree classifier using WEKA.
5. Analyze classification results.
6. Interpret confusion matrices.
7. Compare the performance of different classification algorithms.

Importance of this Experiment:

Classification is one of the most widely used data mining techniques. It is used to predict the category or class of an unknown instance based on historical data. Applications include:

- Placement Prediction
- Disease Diagnosis
- Customer Classification
- Credit Approval
- Email Spam Detection
- Product Recommendation

This experiment introduces students to two popular classification algorithms and helps them understand how predictive models are built and evaluated.

Expected Outcome: After completing this experiment, students will be able to:

1. Build classification models using WEKA.
2. Apply Naive Bayes and J48 classifiers.
3. Interpret classification accuracy.
4. Analyze confusion matrices.

5. Compare classification algorithms.
6. Predict outcomes based on attribute values.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Placement Prediction Dataset (placement_prediction.arff)

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1: Installation and Exploration of WEKA Tool and Available Datasets
- Experiment No. 2: Create and Analyze ARFF Files Using WEKA
- Experiment No. 3: Data Cleaning and Missing Value Treatment
- Experiment No. 4: Data Transformation and Normalization
- Experiment No. 5: Generate Association Rules Using Apriori Algorithm
- Experiment No. 6: Generate Frequent Patterns Using FP-Growth Algorithm and Compare with Apriori
- Attributes and Instances
- Data Preprocessing Concepts

Theory:

1. Classification

Classification is a supervised learning technique used to predict the class label of new data based on previously known examples. A classification model learns from historical data and uses the learned patterns to classify unseen records.

General Process: **Training Data →Model Building→Classification Model→Prediction**

2. Supervised Learning

In supervised learning, the dataset contains a target attribute known as the class attribute.

Example:

Attendance	Marks	Placement
High	Good	Yes
Low	Poor	No

Here: Placement is the class attribute.

3. Naive Bayes Classifier

Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem. It predicts class labels using probability calculations.

Features

- Simple and fast
- Works well on small datasets
- Handles categorical attributes efficiently
- Requires less training time

Advantages

- Easy to implement
- Fast execution
- Effective for classification problems

4. Decision Tree Classification

A Decision Tree classifies data using a set of decision rules arranged in a tree structure. Each internal node represents a decision condition. Each leaf node represents a class label.

5. J48 Decision Tree

J48 is WEKA's implementation of the C4.5 Decision Tree algorithm. It automatically creates decision rules from a dataset.

Example: **Attendance = High →Placement = Yes**

Advantages

- Easy to understand
- Easy to visualize
- Generates human-readable rules

6. Classification Accuracy

Classification accuracy measures how many records are classified correctly.

Formula: **Accuracy = (Correctly Classified Instances / Total Instances) * 100**

Example: 15 Correct Predictions, Accuracy = 15/15 × 100 = 100%

7. Confusion Matrix

A Confusion Matrix is used to evaluate the performance of a classification model. It shows:

- Correct Predictions
- Incorrect Predictions

Example:

Actual / Predicted	Yes	No
Yes	8	0
No	0	7

Industry Insight: Classification techniques are extensively used in healthcare, banking, education, retail analytics, fraud detection, and recommendation systems to support predictive decision-making.

Mini Real-World Example:

Placement Prediction System

Educational institutions often wish to predict whether a student is likely to be placed based on academic and skill-related attributes.

Attributes:

- Attendance
- Marks
- Projects
- Communication Skills

Target: Placement

Values: Yes / No

Classification algorithms can learn from historical placement data and predict future outcomes.

Food Recommendation System

Features:

- Age
- Weight Category
- Nutritional Status
- Food Preference

Target: Recommended Food Category

Classification algorithms can assist in generating suitable food recommendations.

Dataset Used in This Experiment:

Filename: placement_prediction.arff

Dataset Contents:

```
@relation placement
@attribute attendance {High,Low}
@attribute marks {Good,Average,Poor}
@attribute projects {Yes,No}
@attribute communication {Good,Average,Poor}
@attribute placement {Yes,No}
@data
High,Good,Yes,Good,Yes
High,Good,Yes,Average,Yes
High,Average,Yes,Good,Yes
Low,Poor,No,Poor,No
Low,Average,No,Poor,No
High,Good,No,Good,Yes
Low,Poor,Yes,Average,No
High,Average,Yes,Average,Yes
Low,Average,No,Average,No
High,Good,Yes,Good,Yes
Low,Poor,No,Poor,No
High,Average,Yes,Good,Yes
Low,Average,Yes,Average,No
High,Good,No,Average,Yes
Low,Poor,Yes,Poor,No
```

Dataset Information:

- Relation Name: placement
- Instances: 15
- Attributes: 5
- Predictive Attributes: 4
- Class Attribute: placement

Results Obtained During Experiment:

Naive Bayes Results:

Accuracy 100%

Correctly Classified Instances 15

Incorrectly Classified Instances 0

Key Observation

Naive Bayes considered Attendance, Marks, Projects, Communication while creating the classification model.

J48 Results:

Accuracy 100%

Correctly Classified Instances 15

Incorrectly Classified Instances 0

Generated Decision Tree

attendance = High : Yes

attendance = Low : No

Key Observation

J48 identified attendance as the most influential attribute.

Naive Bayes vs J48 Comparison:

Parameter	Naive Bayes	J48
Accuracy	100%	100%
Correctly Classified Instances	15	15
Incorrectly Classified Instances	0	0
Model Type	Probabilistic Classifier	Decision Tree
Main Attribute Used	All Attributes	Attendance

Ease of Interpretation	Moderate	High
Confusion Matrix	Perfect Classification	Perfect Classification

Understanding the Confusion Matrix:

Output

a b <-- classified as

8 0 | a = Yes

0 7 | b = No

Interpretation

Actual Class	Predicted Yes	Predicted No
Yes	8	0
No	0	7

Meaning

8 instances of "Yes" were correctly classified.

7 instances of "No" were correctly classified.

No classification errors occurred.

Accuracy Calculation:

Accuracy = (8 + 7) / 15 × 100 = 15 / 15 × 100 = 100%

Procedure:

Activity 1: Load Dataset

Steps:

1. Open WEKA
2. Select Explorer
3. Open the Preprocess tab
4. Click Open File
5. Select placement_prediction.arff
6. Click Open

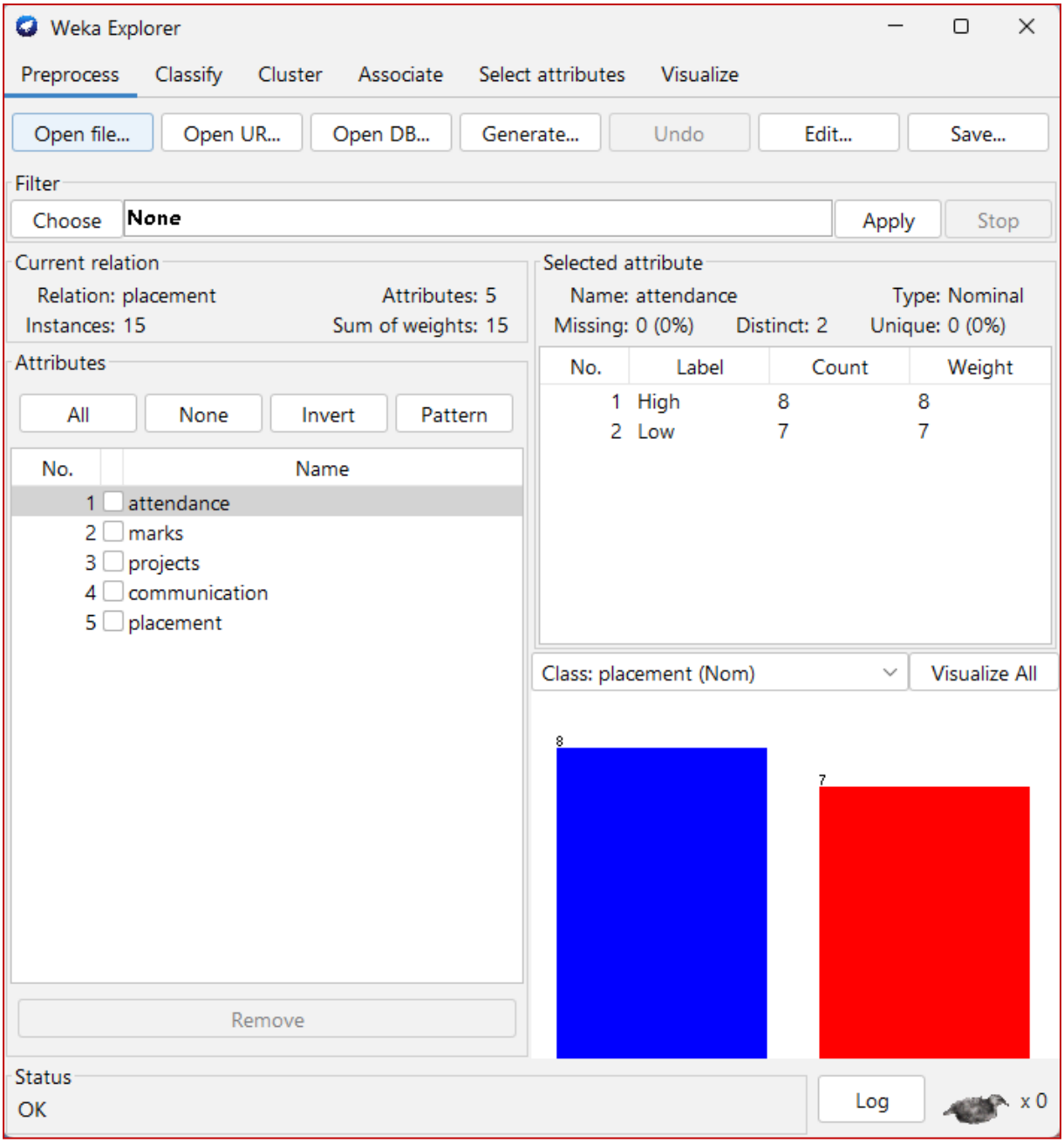


Figure 7.1: Placement Prediction Dataset Loaded in WEKA

Activity 2: Apply Naive Bayes Classifier

Steps:

1. Select: Classify tab
2. Click: Choose
3. Select: bayes → NaiveBayes
4. Select: 10-Fold Cross Validation
5. Click: Start

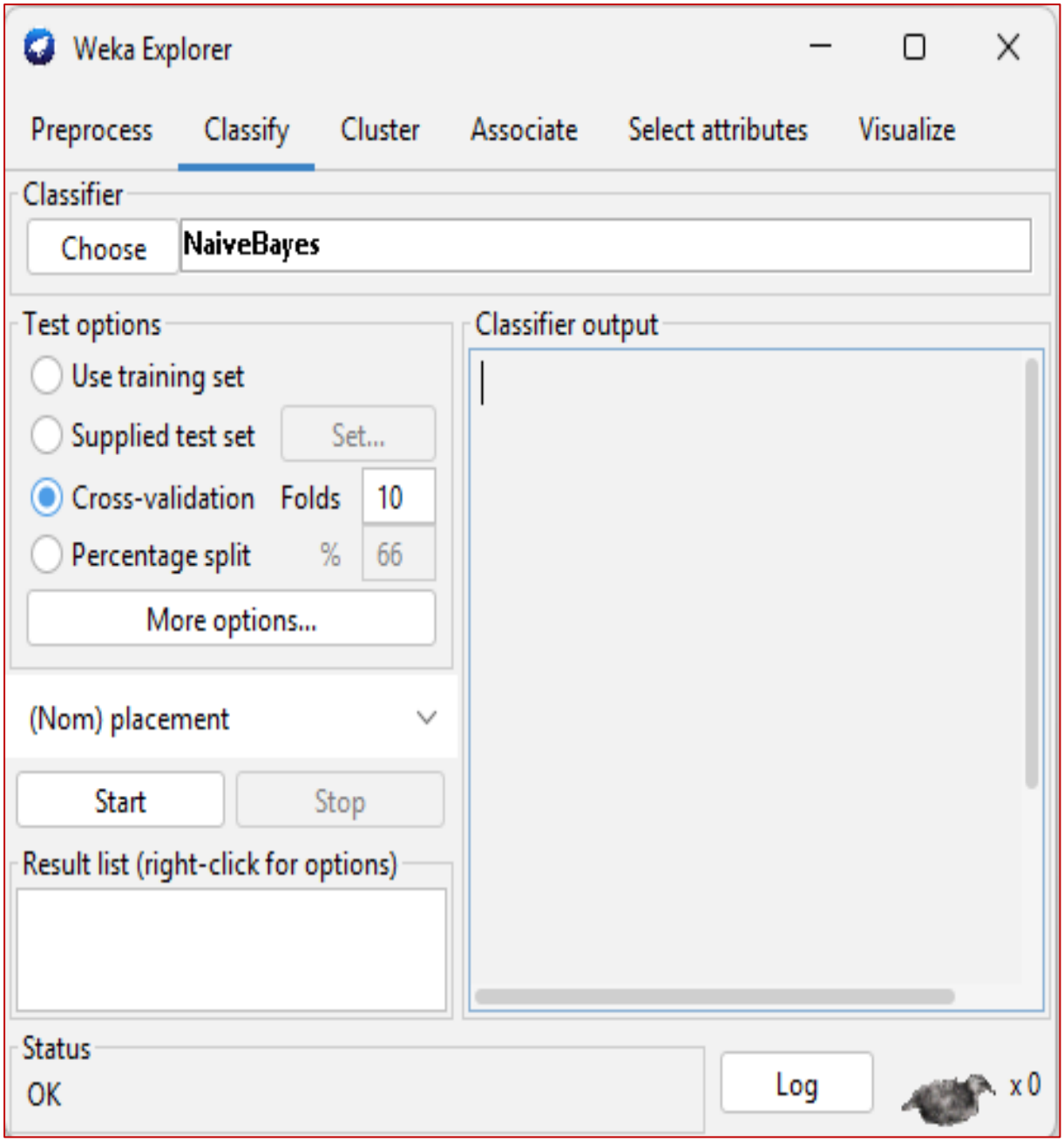


Figure 7.2: Applying Naive Bayes Classifier

Activity 3: Observe Naive Bayes Results

Record:

- Accuracy
- Correctly Classified Instances
- Incorrectly Classified Instances
- Confusion Matrix

```
=== Run information ===

Scheme:
weka.classifiers.bayes.NaiveBayes
Relation:    placement
Instances:   15
Attributes:  5
             attendance
             marks
             projects
             communication
             placement
Test mode:   10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

                Class
Attribute       Yes    No
             (0.53) (0.47)
============================
attendance
  High          9.0   1.0
  Low           1.0   8.0
  [total]      10.0   9.0

marks
  Good          6.0   1.0
  Average       4.0   4.0
  Poor          1.0   5.0
  [total]      11.0  10.0

projects
  Yes           7.0   4.0
  No            3.0   5.0
  [total]      10.0   9.0

communication
  Good          6.0   1.0
  Average       4.0   4.0
  Poor          1.0   5.0
  [total]      11.0  10.0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances          15               100      %
Incorrectly Classified Instances         0                 0      %
Kappa statistic                          1
Mean absolute error                      0.0472
Root mean squared error                  0.0756
Relative absolute error                  9.2859 %
Root relative squared error             14.8353 %
Total Number of Instances               15

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
               1.000    0.000     1.000      1.000    1.000      1.000    1.000     1.000     Yes
               1.000    0.000     1.000      1.000    1.000      1.000    1.000     1.000     No
Weighted Avg.  1.000    0.000     1.000      1.000    1.000      1.000    1.000     1.000

=== Confusion Matrix ===

 a b   <-- classified as
 8 0 | a = Yes
 0 7 | b = No
```

Figure 7.3: Naive Bayes Classification Results

Activity 4: Apply J48 Decision Tree

Steps:

1. Select: Classify tab
2. Click: Choose
3. Select: trees → J48
4. Keep: 10-Fold Cross Validation
5. Click: Start

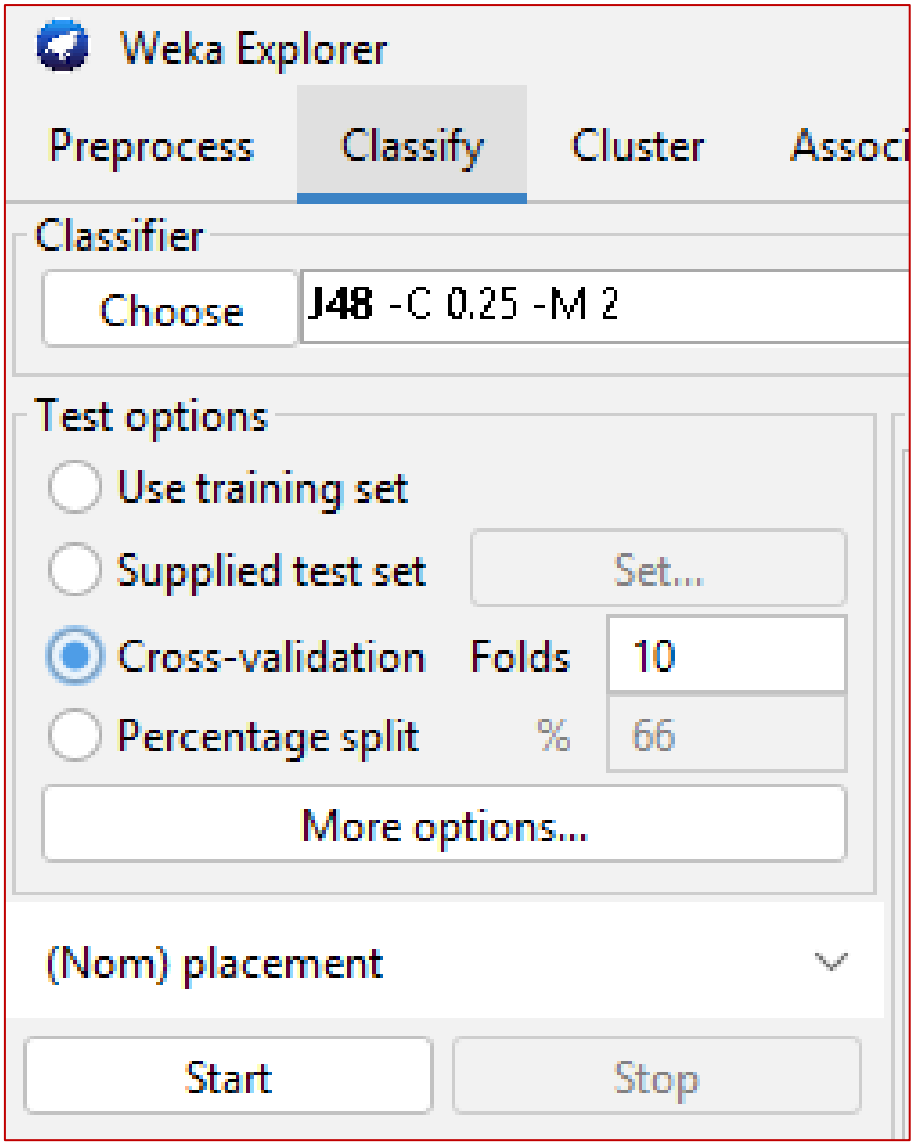


Figure 7.4: Applying J48 Decision Tree

Activity 5: Observe J48 Results

Record:

- Accuracy
- Decision Tree Structure
- Correctly Classified Instances
- Incorrectly Classified Instances
- Confusion Matrix

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:     placement
Instances:    15
Attributes:   5
              attendance
              marks
              projects
              communication
              placement
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree
------------------

attendance = High: Yes (8.0)
attendance = Low: No (7.0)

Number of Leaves  :  2

Size of the tree :  3


Time taken to build model: 0.01 seconds
```

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances          15               100      %
Incorrectly Classified Instances         0                 0      %
Kappa statistic                          1
Mean absolute error                      0
Root mean squared error                  0
Relative absolute error                  0      %
Root relative squared error              0      %
Total Number of Instances               15

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
               1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Yes
               1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     No
Weighted Avg.  1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000

=== Confusion Matrix ===

 a b   <-- classified as
 8 0 | a = Yes
 0 7 | b = No
```

Figure 7.5: J48 Classification Results

Activity 6: Compare Results

Compare:

- Accuracy
- Classification Model
- Confusion Matrix
- Ease of Interpretation

Record observations.

Common Mistakes during Classification

1. Forgetting to select the class attribute.
2. Comparing models using different datasets.
3. Ignoring the confusion matrix.
4. Interpreting accuracy without checking classification errors.
5. Assuming higher accuracy always means a better model.

Common Discussion

1. What is classification?
2. What is supervised learning?
3. What is a class attribute?
4. What is Naive Bayes?
5. What is J48?
6. Why is a confusion matrix important?
7. Which classifier is easier to interpret?
8. Why did J48 select attendance as the root node?
9. What is classification accuracy?
10. Can classification be used in recommendation systems?

Student Activity:

Create a placement prediction dataset containing:

- At least 10 records
- At least 4 predictive attributes
- One class attribute

Apply:

- Naive Bayes
- J48

Compare:

- Accuracy
- Confusion Matrix
- Classification Model

Modify the dataset by:

- Changing attribute values
- Adding new records

Reapply:

- Naïve Bayes
- J48

Observe:

- Changes in Accuracy
- Changes in Decision Tree Structure
- Changes in Confusion Matrix

Observation Table:

PARAMETER	NAIVE BAYES	J48
Accuracy (%)		
Correctly Classified Instances		
Incorrectly Classified Instances		
Main Attribute Used		
Confusion Matrix		
Remarks		

Result: The Naive Bayes and J48 classification algorithms were successfully applied to the placement prediction dataset using WEKA. Both classifiers achieved 100% accuracy under 10-fold cross-validation. The concepts of supervised learning, classification accuracy, decision trees, and confusion matrices were studied successfully.

Viva Questions:

1. What is classification?
2. What is supervised learning?
3. What is a class attribute?
4. What is Naive Bayes?
5. What is Bayes' Theorem?
6. What is a Decision Tree?
7. What is J48?
8. What is classification accuracy?
9. What is a confusion matrix?
10. What is a root node?
11. Which attribute became the root node in this experiment?
12. Differentiate between Naive Bayes and J48.
13. Why is cross-validation used?
14. What is the purpose of 10-fold cross-validation?
15. Give two applications of classification.

Assignment Questions:

1. Explain the Naive Bayes classifier with a suitable example.
2. Explain the J48 Decision Tree algorithm.
3. Compare Naive Bayes and J48.
4. Explain the role of the confusion matrix in classification.
5. Create a classification dataset and compare two classifiers using WEKA.
6. Explain the advantages and limitations of Decision Trees.
7. Discuss applications of classification in educational data mining.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

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